

Every nonzero bounded companion operator on H^∞ fixes a copy of ℓ^∞

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Abstract

For g analytic on the disk, let $S_g f(z) = \int_0^z f'(w)g(w)dw$. Lin, Liu, and Wu ask whether every bounded noncompact $S_g : H^\infty \rightarrow H^\infty$ fixes an isomorphic copy of ℓ^∞ . We prove the stronger statement that every nonzero bounded S_g does so. The copy is produced by derivative interpolation on a geometrically separated sequence along which g is bounded below. This also makes compactness, weak compactness, strict singularity, and the zero symbol equivalent for bounded companion operators on H^∞ .

Status: full affirmative answer, likely valid, pending human review. No later explicit resolution was found in the bounded search described below.

1 Exact source question

For $g \in H(\mathbb{D})$, define

$$(S_g f)(z) = \int_0^z f'(w)g(w)dw, \quad f \in H(\mathbb{D}).$$

Section 5, printed/PDF page 7 of Lin–Liu–Wu, arXiv:1903.10261v3 [1], asks:

Suppose that $S_g : H^\infty \rightarrow H^\infty$ is bounded but not compact. Does S_g fix an isomorphic copy of ℓ^∞ ? In particular, is S_g not strictly singular?

The exact source passage is reproduced here:

consider the trivial equality

$$M_g(f) = f(0)g(0) + T_g(f) + S_g(f) \quad \text{for } z \in \mathbb{D}, f \in H(\mathbb{D}),$$

where M_g is the multiplication operator. It is proven that M_g is not l^2 -singular on H^p while T_g is l^2 -singular on H^p (see [23, 27]). Since the class of l^2 -singular operators forms a linear subspace of the space of all bounded operators, it follows from the above equality that S_g is not l^2 -singular on H^p , that is, $S_g : H^p \rightarrow H^p$ fixes an isomorphic copy of ℓ^2 . The proof is complete. $\square$

5. OPEN QUESTION

In Theorem 1 we exclude the case of $p = \infty$ since our method is not applicable in this case. Although Bourgain [6] has established that a bounded linear operator on H^∞ is weakly compact if and only if it does not fix any copy of ℓ^∞ , we do not know whether or not S_g acting on H^∞ is compact if and only if it does not fix any copy of ℓ^∞ . So we post it as an open question as follows:

Open Question. Suppose that $S_g : H^\infty \rightarrow H^\infty$ is bounded but not compact. Does the operator $S_g : H^\infty \rightarrow H^\infty$ fix an isomorphic copy of ℓ^∞ ? In particular, is the operator S_g not strictly singular?

Anderson–Jovovic–Smith independently prove that the only compact companion operator on H^∞ is the zero operator [2, Proposition 3.3]:

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Indeed, the proof of sufficiency given above is valid for any operator. For an example showing that necessity fails in general, start with the evaluation functional Λ defined on the disk algebra A by $\Lambda f = f(1)$. Then Λ is a norm 1 linear functional on A , and by the Hahn-Banach Theorem can be extended to a linear functional $\hat{\Lambda}$ on H^∞ with $\|\hat{\Lambda}\| = 1$. Let L be the operator on H^∞ that takes $f \in H^\infty$ to the constant function with constant value $\hat{\Lambda}f$. Then L is a rank-one operator, and hence compact. But $L(z^n) = \Lambda(z^n) = 1$ for all positive integers n , and hence condition (ii) fails.

As with boundedness, if two of the operators M_g , S_g and T_g are compact, then so is the third. Since $M_o[H^\infty] = \{0\}$, it is not surprising that the same is true for S_g .

Proposition 3.3. $S_o[H^\infty] = S_o[A] = \{0\}$.

Proof. If S_g is compact on H^∞ , then S_g is bounded and $g \in H^\infty$ by Proposition 2.2. If $g \neq 0$, then without loss of generality we may assume $\|g\|_\infty = 1$. Note that g is not constant, for otherwise S_g would be a constant multiple of a rank-1 perturbation of the identity operator, which is not compact. Thus, the functions g^n , $n = 1, 2, \dots$ converge locally uniformly to 0 in $\mathbb{D}$. We have

$$S_g g^n(z) = \int_0^z g(t)(g^n)'(t) dt = \frac{n}{n+1} g^{n+1}(z) - \frac{n}{n+1} g^{n+1}(0).$$

Since $g^{n+1}(0) \rightarrow 0$, as $n \rightarrow \infty$, but $\|g^{n+1}\|_\infty = 1$ for all n , this violates the condition for compactness Proposition 3.1(ii). Hence $g = 0$, showing that $S_o[H^\infty] = \{0\}$.

The same proof shows that $S_o[A] = \{0\}$, and so will be omitted. $\square$

Although $S_o[H^\infty] = M_o[H^\infty] = \{0\}$, there are non constant functions g for which T_g is compact. We now introduce notation for spaces of functions which we will show have this property. Let

$$H_1^1 = \{g \in H(\mathbb{D}) : g' \in H^1\}$$

so, for example, a conformal map from $\mathbb{D}$ to a domain with rectifiable boundary

The proof below recovers that fact while resolving the stronger fixing question.

2 Proof intuition

The derivative identity

$$(S_g f)' = g f' \tag{1}$$

suggests encoding an arbitrary bounded scalar sequence in normalized derivatives of bounded analytic functions. Choose points z_n tending to the boundary so quickly that they form an H^∞ interpolating sequence, while choosing their arguments so that $|g(z_n)|$ stays uniformly positive. An interpolating Blaschke product forces the constructed functions to vanish at every z_n , and ordinary value interpolation then prescribes their first derivatives. Equation (1) multiplies the encoded coordinate by $g(z_n)$. The Schwarz–Pick derivative bound converts those coordinates back into a lower bound for the H^∞ norm.

3 Three elementary ingredients

We repeatedly use the following consequence of Schwarz–Pick:

$$(1 - |z|^2)|h'(z)| \leq \|h\|_\infty, \quad h \in H^\infty, z \in \mathbb{D}. \tag{2}$$

Lemma 1 (The symbol is bounded). *If $S_g : H^\infty \rightarrow H^\infty$ is bounded, then $g \in H^\infty$ and $\|g\|_\infty \leq \|S_g\|$.*

Proof. Fix $z \in \mathbb{D}$ and let $\phi_z(w) = (z - w)/(1 - \bar{z}w)$. Then $\|\phi_z\|_\infty = 1$ and $|\phi'_z(z)| = (1 - |z|^2)^{-1}$. By (1) and (2),

$$|g(z)| = (1 - |z|^2)|(S_g\phi_z)'(z)| \leq \|S_g\phi_z\|_\infty \leq \|S_g\|.$$

Take the supremum over z . □

Write

$$\rho(z, w) = \left| \frac{z - w}{1 - \bar{w}z} \right|$$

for the pseudohyperbolic distance.

Lemma 2 (Interpolating points where g is large). *If $0 \neq g \in H^\infty$, there are $\delta > 0$ and an interpolating sequence (z_n) for H^∞ such that $|g(z_n)| \geq \delta$ for every n .*

Proof. Choose $a \in \mathbb{D}$ with $g(a) \neq 0$, put $\delta = |g(a)|$, and fix $0 < q < 1$. Choose $\varepsilon > 0$ so small that

$$r_n = 1 - \varepsilon q^n > |a| \quad (n \geq 1).$$

On the circle $|z| = r_n$, choose z_n maximizing $|g|$. The maximum modulus principle on the disk of radius r_n gives $|g(z_n)| \geq |g(a)| = \delta$.

The Blaschke condition holds because $\sum_n (1 - |z_n|) = \varepsilon \sum_n q^n < \infty$. If $m < n$, fixed radii minimize pseudohyperbolic distance when the points have the same argument, so

$$\begin{aligned} \rho(z_m, z_n) &\geq \frac{r_n - r_m}{1 - r_m r_n} \\ &\geq \frac{1 - q^{n-m}}{1 + q^{n-m}}. \end{aligned}$$

The infinite product

$$P_q = \prod_{k=1}^{\infty} \frac{1 - q^k}{1 + q^k}$$

is positive, since the sum of the deficits of its factors is finite. Therefore

$$\inf_n \prod_{m \neq n} \rho(z_m, z_n) \geq P_q^2 > 0.$$

Carleson's interpolation theorem now says that (z_n) is an H^∞ interpolating sequence. □

We use the standard Beurling form of Carleson's theorem: for an interpolating sequence (z_n) there are $u_n \in H^\infty$ such that

$$u_n(z_m) = \delta_{nm}, \quad \sup_{z \in \mathbb{D}} \sum_n |u_n(z)| \leq C. \quad (3)$$

Lemma 3 (A normalized-derivative copy of ℓ^∞). *For the sequence in Lemma 2, there is a bounded linear embedding $J : \ell^\infty \rightarrow H^\infty$ such that*

$$J(c)(z_n) = 0, \quad (1 - |z_n|^2)J(c)'(z_n) = c_n \quad (4)$$

for every $c = (c_n) \in \ell^\infty$.

Proof. Let B be the Blaschke product with simple zeros (z_n) , and put

$$\beta_n = (1 - |z_n|^2)B'(z_n).$$

The standard derivative formula for a Blaschke product gives

$$|\beta_n| = \prod_{m \neq n} \rho(z_m, z_n) \geq P_q^2 =: \eta > 0.$$

Using the Beurling functions in (3), define

$$H_c(z) = \sum_n \frac{c_n}{\beta_n} u_n(z), \quad J(c) = BH_c.$$

This is linear and

$$\|J(c)\|_\infty \leq \|H_c\|_\infty \leq C\eta^{-1}\|c\|_{\ell^\infty}.$$

The interpolation identities give $H_c(z_n) = c_n/\beta_n$, so the product rule yields exactly (4). Conversely, (2) and (4) show

$$\|J(c)\|_\infty \geq \sup_n (1 - |z_n|^2) |J(c)'(z_n)| = \|c\|_{\ell^\infty}.$$

Thus J is an isomorphic embedding and its range is closed. $\square$

4 Full endpoint answer

Theorem 1. *Let $g \in H(\mathbb{D})$ and suppose $S_g : H^\infty \rightarrow H^\infty$ is bounded. If $g \neq 0$, then S_g fixes an isomorphic copy of ℓ^∞ . More precisely, there is an embedding $J : \ell^\infty \rightarrow H^\infty$ and a constant $c_g > 0$ such that*

$$\|S_g J(c)\|_\infty \geq c_g \|J(c)\|_\infty \quad (c \in \ell^\infty).$$

Proof. Lemma 1 gives $g \in H^\infty$. Apply Lemmas 2 and 3. From (1), (4), and (2),

$$\begin{aligned} \|S_g J(c)\|_\infty &\geq \sup_n (1 - |z_n|^2) |(S_g J(c))'(z_n)| \\ &= \sup_n |g(z_n)c_n| \geq \delta \|c\|_{\ell^\infty}. \end{aligned}$$

If $K = C\eta^{-1}$ is the upper embedding constant from Lemma 3, then $\|J(c)\|_\infty \leq K\|c\|_{\ell^\infty}$, and hence the desired estimate holds with $c_g = \delta/K$. Thus the restriction of S_g to the closed subspace $J(\ell^\infty)$ is an isomorphism onto its image. $\square$

Corollary 1. *For every bounded $S_g : H^\infty \rightarrow H^\infty$, the following are equivalent:*

1. $g = 0$;
2. S_g is compact;
3. S_g is weakly compact;
4. S_g is strictly singular;
5. S_g does not fix a copy of ℓ^∞ .

In particular, the open question of [1] has an affirmative answer.

Proof. If $g = 0$, the operator is zero, so all four operator properties follow. If $g \neq 0$, Theorem 1 shows that S_g fixes ℓ^∞ . It is therefore neither compact nor strictly singular. It is not weakly compact either: otherwise its restriction to $J(\ell^\infty)$, followed by the bounded inverse on its range, would make the identity on the nonreflexive space ℓ^∞ weakly compact. $\square$

5 Verification, novelty, and scope

The proof’s quantitative interfaces are visible in the displayed formulas: the geometric radii give uniform Carleson separation; the Blaschke derivative is bounded below by the same separation product; Beurling interpolation provides a bounded linear, not merely pointwise, choice; and Schwarz–Pick turns the prescribed derivative jets into both embedding and operator lower bounds. No compactness characterization is assumed.

The bounded novelty search checked the run’s solution, attempt, proof-gap, and ledger indexes; the parsed local arXiv corpus; exact-title and citation searches; the exact source question; and focused arXiv/web queries combining “Volterra companion”, S_g , H^∞ , weak compactness, strict singularity, and fixing ℓ^∞ . The closest source, Anderson–Jovovic–Smith [2], proves compactness only for the zero symbol but does not state the fixing result. No later explicit answer was found. This is a bounded audit, not a definitive claim of publication novelty.

The result is specific to the companion operator S_g . It does not resolve the distinct symbol-class and compactness questions for $T_g f(z) = \int_0^z f(w)g'(w)dw$. Human review should prioritize the Beurling interpolation statement (3) and the uniform product estimate in Lemma 2.

References

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- [4] J. Bourgain, *New Banach space properties of the disc algebra and H^∞* , *Acta Mathematica* **152** (1984), 1–48.